Assignment 3

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Please note that

- **Release date: March 1, Friday.**
- **Due date: March 13, Wednesday, by 11:55pm.**
- Late submission is **NOT** accepted.
- Please submit your answers as a PDF file with a name like "123010XXX ASS3.pdf" (Your student ID + ASS No.). You may either typeset you answers directly using computers, or scan your handwritten answers. (We recommend you use the printers on campus to scan. If you use your smartphone to scan, please limit the file size 10MB.)

Question 1. Label the following statements as true or false:

(a) If S is a linearly dependent set, then each vector in S is a linear combination of other vectors in S.

(b) Any set containing the zero vector is linearly dependent.

(c) Any nonempty subset of a linearly independent set of vectors $\{\mathbf{v}_1, \ldots, \mathbf{v}_n\}$ is also linearly independent.

(d) Any nonempty subset of a linearly dependent set of vectors $\{\mathbf{v}_1, \ldots, \mathbf{v}_n\}$ is also linearly dependent.

(e) In any vector space, $a\mathbf{x} = b\mathbf{x}$ implies that $a = b$.

(f) In any vector space, $a\mathbf{x} = a\mathbf{y}$ implies that $\mathbf{x} = \mathbf{y}$.

Solution

(a) False. If $\mathbf{x} \neq \mathbf{0}$, then $S = \{\mathbf{x}, \mathbf{0}\}$ is linearly dependent but $\mathbf{x}$ cannot be a linear combination (scalar multiple) of $\mathbf{0}$.

(b) True

(c) True

(d) False. $\{\mathbf{e}_1, \mathbf{e}_2, \mathbf{e}_1 + \mathbf{e}_2\}$ is linearly dependent but the subset $\{\mathbf{e}_1, \mathbf{e}_2\}$ is linearly independent.

(e) False. Take $a = 1$, $b = 2$, $\mathbf{x} = \mathbf{0}$.

(f) False. Take any $\mathbf{x} \neq \mathbf{y}$ and $a = 0$.

Question 2. Determine whether each of the following set of vectors is linearly independent or linearly dependent. If it is linearly dependent, write a non-trivial relation of dependence:

(a)

$$\mathbf{u}_1 = \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \\ 1 \end{bmatrix}, \quad \mathbf{u}_2 = \begin{bmatrix} 1 \\ 2 \\ 3 \\ 4 \\ 5 \end{bmatrix}, \quad \mathbf{u}_3 = \begin{bmatrix} 1 \\ -1 \\ 1 \\ -1 \\ 1 \end{bmatrix}.$$

(b)

$$\mathbf{v}_1 = \begin{bmatrix} 1 \\ -1 \\ 1 \\ -1 \end{bmatrix}, \quad \mathbf{v}_2 = \begin{bmatrix} 1 \\ 1 \\ 2 \\ 3 \end{bmatrix}, \quad \mathbf{v}_3 = \begin{bmatrix} 1 \\ 2 \\ 1 \\ 1 \end{bmatrix}, \quad \mathbf{v}_4 = \begin{bmatrix} 0 \\ 0 \\ 3 \\ 8 \end{bmatrix}.$$

Solution

(a) Let $A = [\mathbf{u}_1, \mathbf{u}_2, \mathbf{u}_3]$. The reduced row echelon form of A is

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix},$$

which implies that the system $A\mathbf{c} = \mathbf{0}$ has only the trivial solution. Hence, the vectors $\mathbf{u}_1, \mathbf{u}_2, \mathbf{u}_3$ are linearly independent.

(b) Let $B = [\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3, \mathbf{v}_4]$. The reduced row echelon form of B is

$$\begin{bmatrix} 1 & 0 & 0 & -1 \\ 0 & 1 & 0 & 3 \\ 0 & 0 & 1 & -2 \\ 0 & 0 & 0 & 0 \end{bmatrix}.$$

So, we see that $\mathbf{c} = (1, -3, 2, 1)^T$ is a non-trivial solution of $A\mathbf{c} = \mathbf{0}$. Hence, $\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3, \mathbf{v}_4$ are linearly dependent and we have a non-trivial

relation of dependence:

$$\mathbf{v}_1 - 3\mathbf{v}_2 + 2\mathbf{v}_3 + \mathbf{v}_4 = \mathbf{0}.$$

Question 3. Let $\mathbf{x}_1, \mathbf{x}_2$, and $\mathbf{x}_3$ be linearly independent vectors in $\mathbb{R}^n$, and let

$$\mathbf{y}_1 = \mathbf{x}_1 + \mathbf{x}_2, \qquad \mathbf{y}_2 = \mathbf{x}_2 + \mathbf{x}_3, \qquad \mathbf{y}_3 = \mathbf{x}_3 + \mathbf{x}_1.$$

Are $\mathbf{y}_1, \mathbf{y}_2, \mathbf{y}_3$ linearly independent? Justify your answer.

Solution

Suppose $c_1\mathbf{y}_1 + c_2\mathbf{y}_2 + c_3\mathbf{y}_3 = \mathbf{0}$. This means that

$$c_1(\mathbf{x}_1 + \mathbf{x}_2) + c_2(\mathbf{x}_2 + \mathbf{x}_3) + c_3(\mathbf{x}_3 + \mathbf{x}_1) = \mathbf{0},$$

which is equivalent to

$$(c_1 + c_3)\mathbf{x}_1 + (c_1 + c_2)\mathbf{x}_2 + (c_2 + c_3)\mathbf{x}_3 = \mathbf{0}.$$

Since $\mathbf{x}_1, \mathbf{x}_2, \mathbf{x}_3$ are linearly independent, their coefficients must all be equal to 0, namely,

$$\begin{cases} c_1 + c_3 = 0 \\ c_1 + c_2 = 0 \\ c_2 + c_3 = 0. \end{cases}$$

The above system has only the trivial solution $c_1 = c_2 = c_3 = 0$. This proves that $\mathbf{y}_1, \mathbf{y}_2, \mathbf{y}_3$ are linearly independent.

Question 4. Let $U = \{\mathbf{u}_1, \ldots, \mathbf{u}_n\}$ be a set of vectors in a vector space V. Prove that U is linearly dependent if and only if $\mathbf{u}_1 = \mathbf{0}$ or $\mathbf{u}_{k+1} \in \text{span}\{\mathbf{u}_1, \ldots, \mathbf{u}_k\}$ for some k $(1 \leq k < n)$.

Solution

($\Leftarrow$). If $\mathbf{u}_1 = \mathbf{0}$, then we have a non-trivial dependence relation:

$$1\mathbf{u}_1 + 0\mathbf{u}_2 + \cdots + 0\mathbf{u}_n = \mathbf{0}.$$

If $\mathbf{u}_{k+1} \in \text{span}\{\mathbf{u}_1, \ldots, \mathbf{u}_k\}$ for some k $(1 \leq k < n)$, then there exist $c_1, \ldots, c_k \in \mathbb{R}$ such that $\mathbf{u}_{k+1} = c_1\mathbf{u}_1 + \cdots + c_k\mathbf{u}_k$, so we also have a non-trivial dependence relation:

$$c_1\mathbf{u}_1 + \cdots + c_k\mathbf{u}_k - 1\mathbf{u}_{k+1} + 0\mathbf{u}_{k+2} + \cdots + 0\mathbf{u}_n = \mathbf{0}.$$

Hence, U is linearly dependent.
($\Rightarrow$). Suppose U is linearly dependent. Then there exist scalars $c_1, \ldots, c_n \in \mathbb{R}$ which are not all 0 such that

$$c_1\mathbf{u}_1 + c_2\mathbf{u}_2 + \cdots + c_n\mathbf{u}_n = \mathbf{0}. \qquad (*)$$

We will prove the result by contradiction, so let us assume that

$$\mathbf{u}_1 \neq \mathbf{0} \quad \text{and} \quad \mathbf{u}_{k+1} \notin \operatorname{span}\{\mathbf{u}_1, \ldots, \mathbf{u}_k\} \text{ for all } k \text{ with } 1 \leq k < n. \qquad (\dagger)$$

First, note that c_n must be 0, because if not, then from $(*)$ we could write

$$\mathbf{u}_n = -\frac{c_1}{c_n}\mathbf{u}_1 - \frac{c_2}{c_n}\mathbf{u}_2 - \cdots - \frac{c_{n-1}}{c_n}\mathbf{u}_{n-1},$$

which is impossible since $\mathbf{u}_n \notin \operatorname{span}\{\mathbf{u}_1, \ldots, \mathbf{u}_{n-1}\}$ according to $(\dagger)$. So $c_n = 0$ and $(*)$ becomes

$$c_1\mathbf{u}_1 + c_2\mathbf{u}_2 + \cdots + c_{n-1}\mathbf{u}_{n-1} = \mathbf{0}.$$

By the same reasoning, we deduce that c_{n-1} must also be 0 because $\mathbf{u}_{n-1} \notin \operatorname{span}\{\mathbf{u}_1, \ldots, \mathbf{u}_{n-2}\}$ according to $(\dagger)$.
Inductively, we deduce that $c_n = c_{n-1} = \cdots = c_2 = 0$ and thus $(*)$ simplifies to

$$c_1\mathbf{u}_1 = \mathbf{0}.$$

But by $(\dagger)$, $\mathbf{u}_1 \neq \mathbf{0}$, so we must have $c_1 = 0$. We have deduced that

$$c_n = c_{n-1} = \cdots = c_2 = c_1 = 0,$$

which contradicts the fact from the beginning that the scalars $c_1, \ldots, c_n$ are not all 0. Therefore, we conclude that $(\dagger)$ must be false and hence we have $\mathbf{u}_1 = \mathbf{0}$ or $\mathbf{u}_{k+1} \in \operatorname{span}\{\mathbf{u}_1, \ldots, \mathbf{u}_k\}$ for some k $(1 \leq k < n)$.

Question 5. Let U_1 and U_2 be two finite sets of vectors in a vector space V. Show that if $U_1 \subseteq U_2$, then $\operatorname{span}(U_1) \subseteq \operatorname{span}(U_2)$. In particular, show that if $U_1 \subseteq U_2$ and U_1 is a spanning set of V, then U_2 is also a spanning set of V.

Solution

Suppose $U_1 \subseteq U_2$. Then we can write

$$U_1 = \{\mathbf{u}_1, \ldots, \mathbf{u}_n\}, \quad U_2 = \{\mathbf{u}_1, \ldots, \mathbf{u}_n, \mathbf{u}_{n+1}, \ldots, \mathbf{u}_{n+m}\},$$

where $m \geq n$. For any $\mathbf{x} \in \text{span}(U_1)$, there exist $c_1, \ldots, c_n \in \mathbb{R}$ such that

$$\begin{aligned}\mathbf{x} &= c_1\mathbf{u}_1 + \cdots + c_n\mathbf{u}_n \\ &= c_1\mathbf{u}_1 + \cdots + c_n\mathbf{u}_n + 0\mathbf{u}_{n+1} + \cdots + 0\mathbf{u}_{n+m} \in \text{span}(U_2).\end{aligned}$$

Hence $\text{span}(U_1) \subseteq \text{span}(U_2)$. If, moreover, U_1 is a spanning set of V, i.e., $\text{span}(U_1) = V$, then

$$V = \text{span}(U_1) \subseteq \text{span}(U_2) \subseteq V.$$

This implies that $\text{span}(U_2) = V$, which means that U_2 is a spanning set of V.

Question 6. Let

$$\mathbf{x}_1 = \begin{bmatrix} -1 \\ 2 \\ 3 \end{bmatrix}, \quad \mathbf{x}_2 = \begin{bmatrix} 3 \\ 4 \\ 2 \end{bmatrix}, \quad \mathbf{x} = \begin{bmatrix} 2 \\ 6 \\ 6 \end{bmatrix}, \quad \mathbf{y} = \begin{bmatrix} -9 \\ -2 \\ 5 \end{bmatrix}.$$

(a) Is $\mathbf{x} \in \text{span}\{\mathbf{x}_1, \mathbf{x}_2\}$?

(b) Is $\mathbf{y} \in \text{span}\{\mathbf{x}_1, \mathbf{x}_2\}$?

Justify your answers.

Solution

(a) Consider $c_1\mathbf{x}_1 + c_2\mathbf{x}_2 = \mathbf{x}$. This is equivalent to solving the linear system

$$\left[\begin{array}{cc|c} -1 & 3 & 2 \\ 2 & 4 & 6 \\ 3 & 2 & 6 \end{array}\right] \to \left[\begin{array}{cc|c} -1 & 3 & 2 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{array}\right],$$

which implies that the system has no solution. Hence $\mathbf{x} \notin \text{span}\{\mathbf{x}_1, \mathbf{x}_2\}$.

(b) Consider $c_1\mathbf{x}_1 + c_2\mathbf{x}_2 = \mathbf{y}$. This is equivalent to solving the linear system

$$\left[\begin{array}{cc|c} -1 & 3 & -9 \\ 2 & 4 & -2 \\ 3 & 2 & 5 \end{array}\right] \to \left[\begin{array}{cc|c} -1 & 3 & 9 \\ 0 & 10 & -20 \\ 0 & 0 & 0 \end{array}\right],$$

which implies that the system has a solution. Hence $\mathbf{y} \in \text{span}\{\mathbf{x}_1, \mathbf{x}_2\}$.

Question 7. Let V be a vector space over $\mathbb{R}$. Use the axioms of vector space to prove the following statements:

(a) $a\mathbf{0} = \mathbf{0}$ for any $a \in \mathbb{R}$.

(b) For any $\mathbf{x}, \mathbf{y} \in V$, if $\mathbf{x} + \mathbf{y} = \mathbf{0}$, then $\mathbf{y} = -\mathbf{x}$.

(c) $-(a\mathbf{x}) = (-a)\mathbf{x}$ for any $\mathbf{x} \in V$ and $a \in \mathbb{R}$.

Solution

(a)

$$\begin{aligned}
a\mathbf{0} &= a\mathbf{0} + \mathbf{0} && (A3)\\
&= a\mathbf{0} + (a\mathbf{0} + (-a\mathbf{0})) && (A4)\\
&= (a\mathbf{0} + a\mathbf{0}) + (-a\mathbf{0}) && (A2)\\
&= a(\mathbf{0} + \mathbf{0}) + (-a\mathbf{0}) && (A5)\\
&= a\mathbf{0} + (-a\mathbf{0}) && (A3)\\
&= \mathbf{0} && (A4)
\end{aligned}$$

(b)

$$\begin{aligned}
\mathbf{y} &= \mathbf{y} + \mathbf{0} && (A3)\\
&= \mathbf{y} + (\mathbf{x} + (-\mathbf{x})) && (A4)\\
&= (\mathbf{y} + \mathbf{x}) + (-\mathbf{x}) && (A2)\\
&= (\mathbf{x} + \mathbf{y}) + (-\mathbf{x}) && (A1)\\
&= \mathbf{0} + (-\mathbf{x}) && (\mathbf{x} + \mathbf{y} = \mathbf{0})\\
&= (-\mathbf{x}) + \mathbf{0} && (A1)\\
&= -\mathbf{x} && (A4)
\end{aligned}$$

(c) By part (b), we only need to show that $a\mathbf{x} + (-a)\mathbf{x} = \mathbf{0}$.

$$\begin{aligned}
a\mathbf{x} + (-a)\mathbf{x} &= (a + (-a))\mathbf{x} && (A6)\\
&= 0\mathbf{x} && (a + (-a) = 0)\\
&= 0\mathbf{x} + (\mathbf{x} + (-\mathbf{x})) && (A3, A4)\\
&= (0 + 1)\mathbf{x} + (-\mathbf{x}) && (A8, A2, A6)\\
&= \mathbf{x} + (-\mathbf{x}) && (A8)\\
&= \mathbf{0} && (A4)
\end{aligned}$$

Question 8. Let V be a vector space. Use the axioms of vector space to prove the cancellation law: for any $\mathbf{x}, \mathbf{y}, \mathbf{z} \in V$, if $\mathbf{x} + \mathbf{y} = \mathbf{x} + \mathbf{z}$, then $\mathbf{y} = \mathbf{z}$.

Solution

Suppose $\mathbf{x}+\mathbf{y}=\mathbf{x}+\mathbf{z}$. Then by (A1),

$$\mathbf{y}+\mathbf{x}=\mathbf{x}+\mathbf{y}=\mathbf{x}+\mathbf{z}=\mathbf{z}+\mathbf{x}.$$

Then

$$\begin{aligned}
\mathbf{y} &= \mathbf{y}+\mathbf{0} && (A3)\\
&= \mathbf{y}+(\mathbf{x}+(-\mathbf{x})) && (A4)\\
&= (\mathbf{y}+\mathbf{x})+(-\mathbf{x}) && (A2)\\
&= (\mathbf{z}+\mathbf{x})+(-\mathbf{x}) && (\mathbf{y}+\mathbf{x}=\mathbf{z}+\mathbf{x})\\
&= \mathbf{z}+(\mathbf{x}+(-\mathbf{x})) && (A2)\\
&= \mathbf{z}+\mathbf{0} && (A4)\\
&= \mathbf{z} && (A3)
\end{aligned}$$

Question 9. Let $\mathbb{R}^+$ denote the set of all positive real numbers. Define the scalar multiplication operation, denoted $\circ$, by

$$\alpha \circ x = x^\alpha \qquad \text{for all } x \in \mathbb{R}^+, \alpha \in \mathbb{R}.$$

Define the addition operation, denoted $\oplus$, by

$$x \oplus y = x \cdot y \qquad \text{for all } x, y \in \mathbb{R}^+.$$

For example, we have

$$-2 \circ \frac{1}{3} = \left(\frac{1}{3}\right)^{-2} = 9 \quad \text{and} \quad 4 \oplus 5 = 4 \cdot 5 = 20.$$

Is $\mathbb{R}^+$ a vector space with these operations? Justify your answer.

Solution

Yes. $\mathbb{R}^+$ is closed under addition $\oplus$ and scalar multiplication $\circ$ defined above, and the right axioms of vector space can be verified:

(A1) For any $x, y \in \mathbb{R}^+$, $x \oplus y = x \cdot y = y \cdot x = y \oplus x$.

(A2) For any $x, y, z \in \mathbb{R}^+$, $(x \oplus y) \oplus z = x \cdot y \cdot z = x \oplus (y \oplus z)$.

(A3) Define 1 to be the "zero vector" in $\mathbb{R}^+$. Then $x \oplus 1 = x \cdot 1 = x$ for all $x \in \mathbb{R}^+$.

(A4) For any $x \in \mathbb{R}^+$, define $-x = \frac{1}{x}$. Then $x \oplus (-x) = x \cdot \frac{1}{x} = 1$, which is

the "zero vector" in $\mathbb{R}^+$.

(A5) For any $\alpha \in \mathbb{R}$ and $x, y \in \mathbb{R}^+$,

$$\alpha \circ (x \oplus y) = (x \oplus y)^\alpha = (x \cdot y)^\alpha = x^\alpha \cdot y^\alpha = (\alpha \circ x) \oplus (\alpha \circ y).$$

(A6) For any $\alpha, \beta \in \mathbb{R}$ and $x \in \mathbb{R}^+$,

$$(\alpha + \beta) \circ x = x^{\alpha+\beta} = x^\alpha \cdot x^\beta = (\alpha \circ x) \oplus (\beta \circ x).$$

(A7) For any $\alpha, \beta \in \mathbb{R}$ and $x \in \mathbb{R}^+$,

$$\alpha \circ (\beta \circ x) = (\beta \circ x)^\alpha = (x^\beta)^\alpha = x^{\alpha\beta} = (\alpha\beta) \circ x.$$

(A8) For any $x \in \mathbb{R}^+$, $1 \circ x = x^1 = x$.

Therefore, $\mathbb{R}^+$ is a vector space.

Question 10. Let $V = \{(a_1, a_2) : a_1, a_2 \in \mathbb{R}\}$. For $(a_1, a_2), (b_1, b_2) \in V$ and $c \in \mathbb{R}$, define addition and scalar multiplication by

$$(a_1, a_2) + (b_1, b_2) = (a_1 + 2b_1, a_2 + 3b_2) \quad \text{and} \quad c(a_1, a_2) = (ca_1, ca_2).$$

Is V a vector space with these operations? Justify your answer.

Solution

No, because axiom (A1) does not hold:

$$(1, 0) + (0, 1) = (1, 3) \quad \text{but} \quad (0, 1) + (1, 0) = (2, 1) \neq (1, 0) + (0, 1).$$

Question 11. Let U and V be subspaces of a vector space W. Define

$$U + V = \{\mathbf{u} + \mathbf{v} \,|\, \mathbf{u} \in U \text{ and } \mathbf{v} \in V\}.$$

Show that $U + V$ is a subspace of W.

Solution

(i) Since U and V are both subspaces, they both contain the zero vector $\mathbf{0}$, and we have $\mathbf{0} = \mathbf{0} + \mathbf{0}$, so $\mathbf{0} \in U + V$.

(ii) If $\mathbf{x}, \mathbf{y} \in U + V$, then

$$\begin{cases} \mathbf{x} = \mathbf{u} + \mathbf{v} \\ \mathbf{y} = \mathbf{u}' + \mathbf{v}' \end{cases}$$

for some $\mathbf{u}, \mathbf{u}' \in U$ and $\mathbf{v}, \mathbf{v}' \in V$. Then

$$\mathbf{x} + \mathbf{y} = (\mathbf{u} + \mathbf{v}) + (\mathbf{u}' + \mathbf{v}') = (\mathbf{u} + \mathbf{u}') + (\mathbf{v} + \mathbf{v}').$$

Note that $\mathbf{u} + \mathbf{u}' \in U$ because U is a subspace. Similarly, $\mathbf{v} + \mathbf{v}' \in V$ because V is a subspace. This shows that $\mathbf{x} + \mathbf{y} \in U + V$.

(iii) If $\mathbf{x} \in U + V$ and $\alpha \in \mathbb{R}$, then

$$\mathbf{x} = \mathbf{u} + \mathbf{v}$$

for some $\mathbf{u} \in U$ and $\mathbf{v} \in V$. Then

$$\alpha\mathbf{x} = \alpha(\mathbf{u} + \mathbf{v}) = (\alpha\mathbf{u}) + (\alpha\mathbf{v}).$$

Note that $\alpha\mathbf{u} \in U$ and $\alpha\mathbf{v} \in V$ because U and V are subspaces. Hence, $\alpha\mathbf{x} \in U + V$.

Therefore, $U + V$ is a subspace of W.

Question 12. Determine whether the following sets form subspaces of $\mathbb{R}^3$:

(a) $\{(x_1, x_2, x_3)^T \,|\, x_1 + x_3 = 1\}$

(b) $\{(x_1, x_2, x_3)^T \,|\, x_1 = x_2 = x_3\}$

Solution

(a). No, because the set does not contain the zero vector $\mathbf{0} = (0, 0, 0)^T$.

(b). Let $W = \{\mathbf{x} = (x_1, x_2, x_3)^T \,|\, x_1 = x_2 = x_3\}$.

(i) $\mathbf{0} \in W$ since $0 = 0 = 0$.

(ii) If $\mathbf{x}, \mathbf{y} \in W$, namely,

$$\begin{cases} x_1 = x_2 = x_3 \\ y_1 = y_2 = y_3, \end{cases}$$

then adding the two equations yields

$$x_1 + y_1 = x_2 + y_2 = x_3 + y_3,$$

which means that $\mathbf{x} + \mathbf{y} \in W$.

(iii) If $\alpha \in \mathbb{R}$ and $\mathbf{x} \in W$, then

$$x_1 = x_2 = x_3,$$

which implies

$$\alpha x_1 = \alpha x_2 = \alpha x_3,$$

and hence $\alpha\mathbf{x} \in W$.

Therefore, W is a subspace of $\mathbb{R}^3$.

Question 13. Let

$$A_1 = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}, \quad A_2 = \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix}, \quad A_3 = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}.$$

Show that $\text{span}\{A_1, A_2, A_3\}$ is the set of all 2×2 symmetric matrices.

Solution

$$\begin{aligned} \text{span}\{A_1, A_2, A_3\} &= \left\{ a \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix} + b \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix} + c \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} : a, b, c \in \mathbb{R} \right\} \\ &= \left\{ \begin{bmatrix} a & c \\ c & b \end{bmatrix} : a, b, c \in \mathbb{R} \right\}, \end{aligned}$$

which is equal to the set of all 2×2 symmetric matrices.

Question 14. Which of the following are subspaces of $\mathbb{R}^{n \times n}$? Justify your answers.

(a) The set of all $n \times n$ upper triangular matrices

(b) The set of all $n \times n$ skew-symmetric matrices

(c) The set of all $n \times n$ nonsingular matrices

Solution

(a) Yes.

(i) The zero matrix O is upper triangular.

(ii) If A, B are upper triangular, then $A + B$ is also upper triangular.

(iii) If A is upper triangular and $\alpha \in \mathbb{R}$, then αA is also upper triangular.

(b) Yes. Recall that A is skew-symmetric if $A^T = -A$.

(i) O is skew-symmetric since $O^T = O = -O$.

(ii) If A, B are skew-symmetric, then $(A+B)^T = A^T + B^T = -A - B = -(A+B)$, so $A+B$ is also skew-symmetric.

(iii) If A is skew-symmetric and $\alpha \in \mathbb{R}$, then $(\alpha A)^T = \alpha A^T = \alpha(-A) = -(\alpha A)$, so αA is also skew-symmetric.

(c) No. The set does not contain the zero matrix O, which is not invertible.

Question 15. Is $\{f \in C[a,b] \mid f(a) = f(b)\}$ is a subspace of $C[a,b]$? Justify your answer.

Solution

Denote $W = \{f \in C[a,b] \mid f(a) = f(b)\}$.

(i) The zero function $\mathbf{0}$ satisfies $\mathbf{0}(a) = 0 = \mathbf{0}(b)$, so $\mathbf{0} \in W$.

(ii) If $f, g \in W$, then $f(a) = f(b)$ and $g(a) = g(b)$, which imply that

$$f(a) + g(a) = f(b) + g(b)$$

and hence $f + g \in W$.

(iii) If $\alpha \in \mathbb{R}$ and $f \in W$, then $f(a) = f(b)$ which implies

$$\alpha f(a) = \alpha f(b),$$

so $\alpha f \in W$.

Therefore, W is a subspace of $C[a,b]$.